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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.2

$$x^2 - 4 = (x-2)(x+2)$$

$$\frac{1}{x^2-4} = \frac{1}{(x-2)(x+2)} = \frac{A}{x-2} + \frac{B}{x+2}$$

$$A = \left.\frac{1}{x+2}\right|_{x=2} = \frac{1}{4}$$

$$B = \left.\frac{1}{x-2}\right|_{x=-2} = -\frac{1}{4}$$

$$\frac{1}{x^2-4} = \frac{1}{4(x-2)} - \frac{1}{4(x+2)}$$

$$\int \frac{dx}{x^2-4} = \frac{1}{4}\int \frac{dx}{x-2} - \frac{1}{4}\int \frac{dx}{x+2}$$

$$= \frac{1}{4}\ln|x-2| - \frac{1}{4}\ln|x+2| + c$$

$$= \frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + c$$

References